Stormwater and Urban Water Systems Modeling Conference

QUANTIFYING THE STORMWATER POLLUTION REDUCTION BENEFITS ASSOCIATED WITH TRADITIONAL PUBLIC WORKS MAINTENANCE PRACTICES

Presented by

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In 1999, Livonia, Michigan authorized HRC and PWR to conduct a:

Pollutant Reduction Study of Improved Maintenance Practices Applied to Livonia's Portion of the Bell Branch and Tarabusi Creek Subwatershed

Partially funded by the
Rouge River National Wet Weather Demonstration Project

Study Motivation

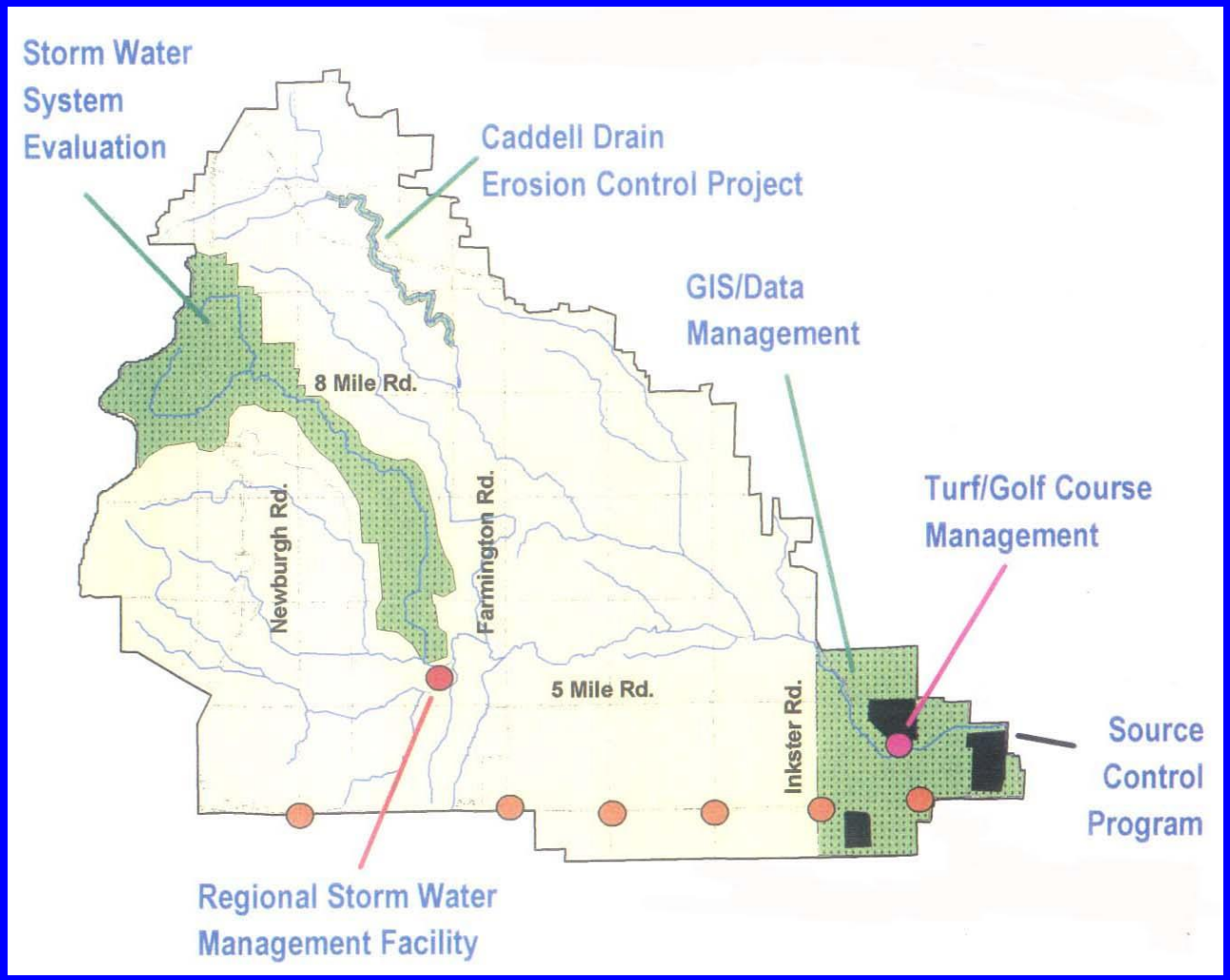
In 1997, a Management Study recommended that local jurisdictions conduct:

Storm Water System Maintenance Activities that reduce or eliminate pollutant loadings in storm water at the source.

Maintenance Activities include:

- Street Sweeping
- Catchbasin Cleaning
- Leaf Collection

Watershed Map



Problem:

Most storm water studies can not afford continuously monitoring the quantity and quality of storm water events from small homogenous sites.

Solution:

Representative land use sites can be selected and monitored for the accumulation of sediments and associated pollutants at a fraction of the cost.

Livonia Study Design

- Delineate watershed land use characteristics
  - use best available mapping
  - conduct “windshield surveys”
- Select land use monitoring site
- Periodically monitor sediment accumulations
  - street and parking lot surfaces
  - trapped within catchbasins
- Periodically conduct physical and chemical analyses
  - sieve into eight particle size fractions
  - composite back to three fractions for chemical analysis of Cl, COD, TP, OrthoP, ten metals and SPLP

Representative Single-Family Residential

Representative Commercial Parking Site

Representative Recreational Parking Site

Sediment sampling at accumulation monitoring sites.

Livonia Study Design (cont):

- Obtain nearby hourly rainfall data
  - entire monitoring period for SIMPTM calibration
  - long historic record for average rainfall year creation
- Calibrate SIMPTM
  - match simulated versus observed sediment accumulations
- Estimate unit costs of maintenance activities
  - $ per cub kilometer or hectares swept
  - $ per catchbasin cleaned
- Conduct Alternative BMP Evaluation
  - use average rainfall year as simulation period
  - use unit cost data to estimate optimal levels of effort
  - use chemical results to simulate pollutant loadings

SIMplified Particulate Transport Model (SIMPTM)

- Continuous urban storm water quality model
- Accurately simulates the accumulation and washoff of sediment and its associated pollutants
- Simulates storm water pollutant reductions associated with street sweeping and catchbasin cleaning
- See Volume 4, Chapter 1; Volume 5, Chapter 9, and Volume 6, Chapter 18 of Toronto Conference Series for more information on SIMPTM and its use

SIMPTM Calibration

- Used hourly data from SEMCOG gauge within 2.2 km of the monitoring sites
- 52 rainfall runoff events totaling 418 mm
- Used sieve analysis results to assign particle size distribution of incrementally accumulating sediments
- Set most model basin parameters on physical site characteristics and previous washoff calibrations
- Varied accumulation parameters to attempt to match observed sediment accumulations on paved surfaces and within catchbasins

Observed Versus Simulated Sediment Accumulations

Single-Family Residential Street Sites

Land Use/Site Name	Test Area	Sampling Data	Observed Accumulation kg/ha	Simulated Accumulation kg/ha
Single-Family Residential Street at Munger	15-G	9/10/99	152	152
	15-G	11/4/99	26	107
	15-G	1/6/00	48	74
	15-G	3/24/00	23	39
	15-P	3/24/00	32	42
	15-G	5/16/00	314	87
	15-P	5/16/00	23	87
Single-Family Residential Street at Riverside	20-G	9/10/99	40	40
	20-G	11/4/99	44	51
	20-G	1/6/00	45	71
	20-G	3/24/00	29	34
	20-P	3/24/00	66	39
	20-G	5/22/00	126	99
	20-P	5/22/00	40	99

Observed Versus Simulated Sediment Accumulations

Parking Lot Sites

Land Use/Site Name	Test Area	Sampling Data	Observed Accumulation kg/ha	Simulated Accumulation kg/ha
Shopping Center Parking at Newburgh	1-G	9/10/99	209	214
	1-G	11/4/99	151	179
	1-G	1/6/00	874	-
	1-G	3/24/00	127	194
	1-P	3/24/00	447	447
	1-P	5/16/00	192	200
Other Area Parking at Fox Creek Golf Course	2A-G	9/10/99	210	210
	2A-G	11/4/99	109	134
	2A-G	1/6/00	152	158
	2A-G	3/24/00	138	120
	2A-P	3/24/00	296	296
	2A-P	5/16/00	350	172

Observed Versus Simulated Catchbasin Accumulations

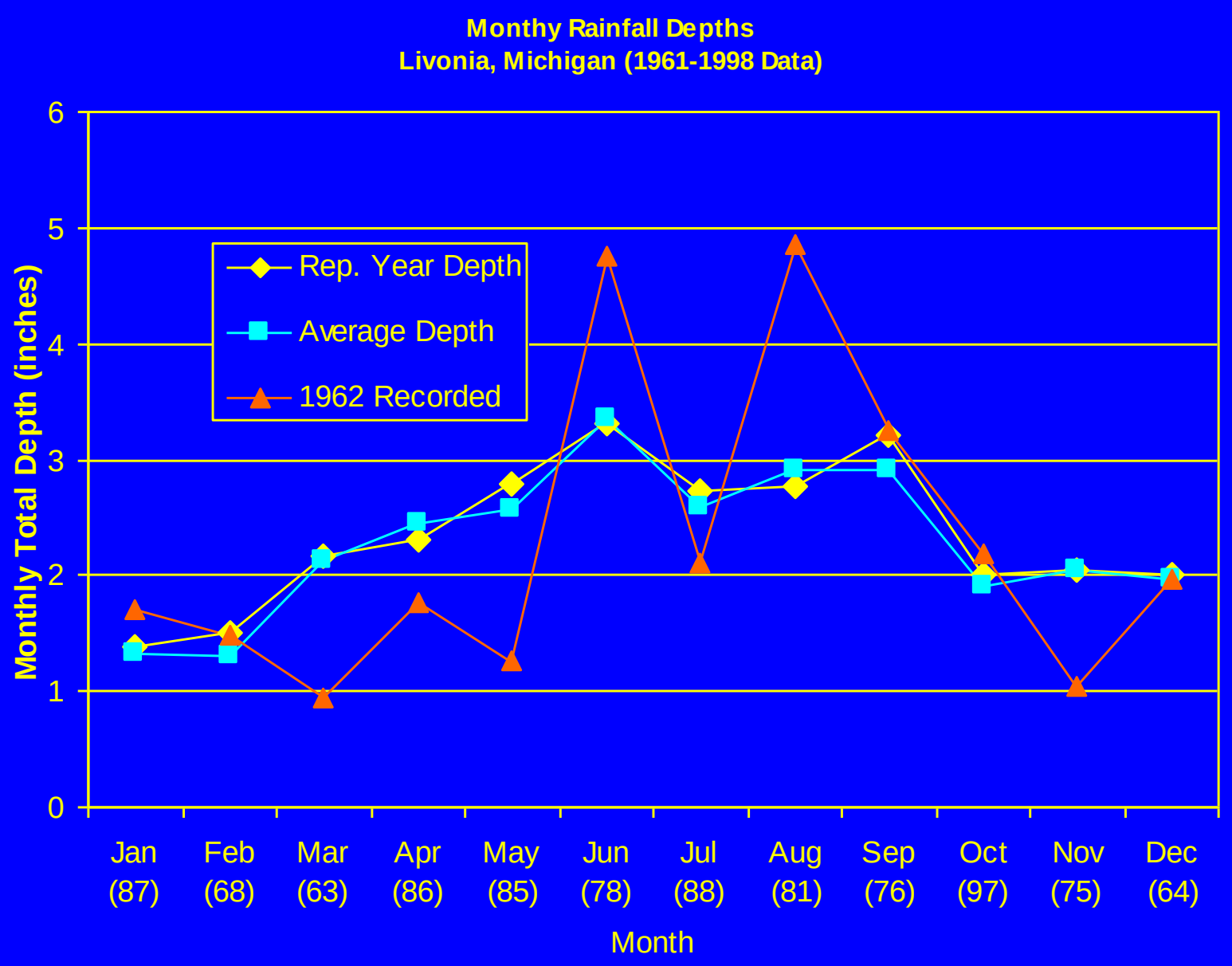
Site Name	Monitoring Date	No. of Catchbasins	Observed Accumulation Avg. Depth of Sediment (m)	Simulated Accumulation Avg. Depth of Sediment (m)
Newburgh	5/11/00	7	.018	.006
Fox Creek	3/24/00	8	.012	.015
Munger	5/11/00	8	.015	.006
Riverside	3/24/00	14	.009	.003

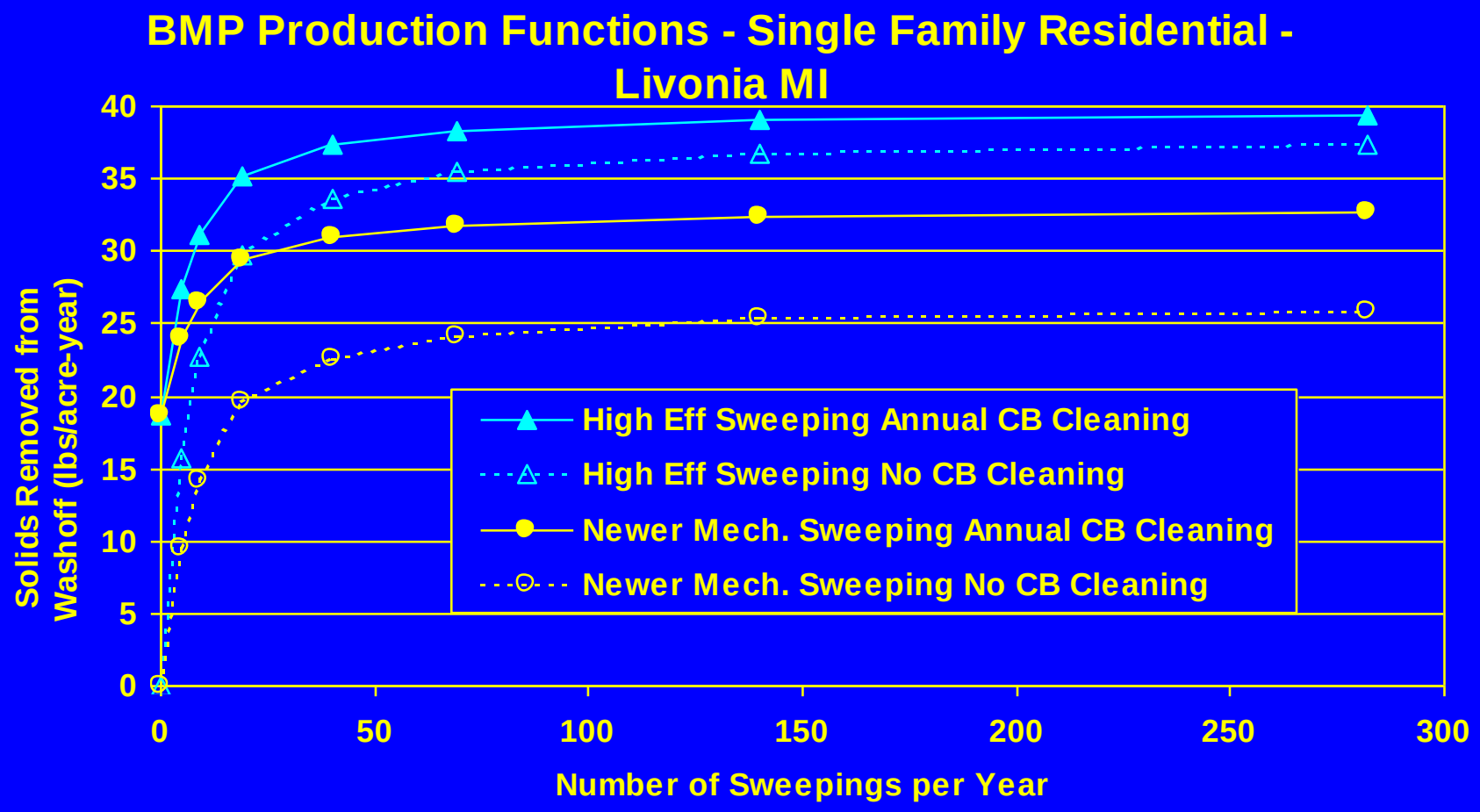
Unit Costs of Maintenance Activities

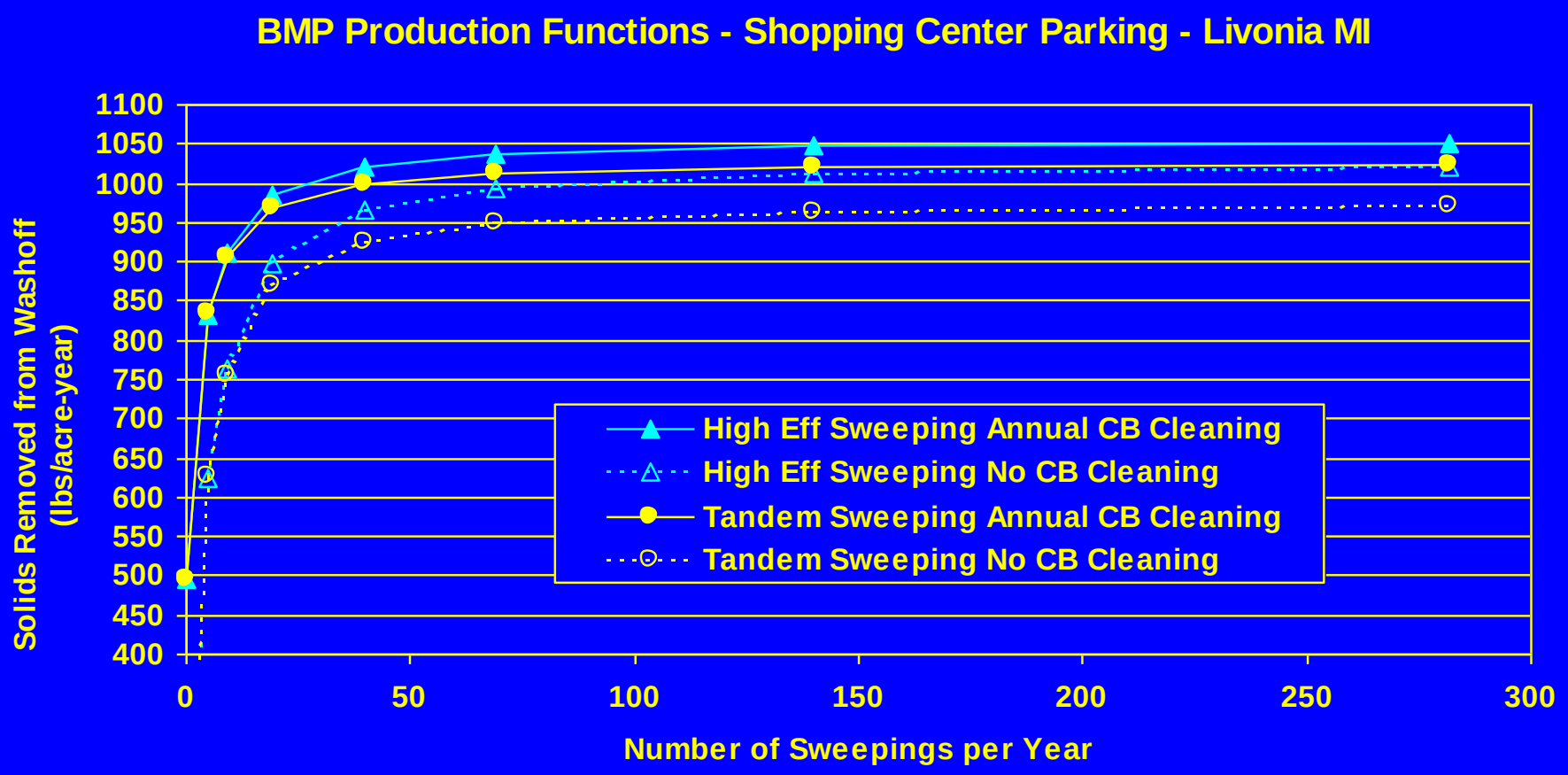
- April to November City sweeps residential 4 times/year and major roads 7 to 8 times with either a mechanical or regenerative air sweeper
- Annually sweeps 1193 curb km at $123/curb km swept
- City cleans publicly-owned catchbasins every 2.5 years
- Annually cleans 3290 catchbasins at $44.25/unit cleaned
- City does not sweep or clean privately-owned properties or publicly-owned parking lots
- Costs include labor, overtime, equipment and overhead

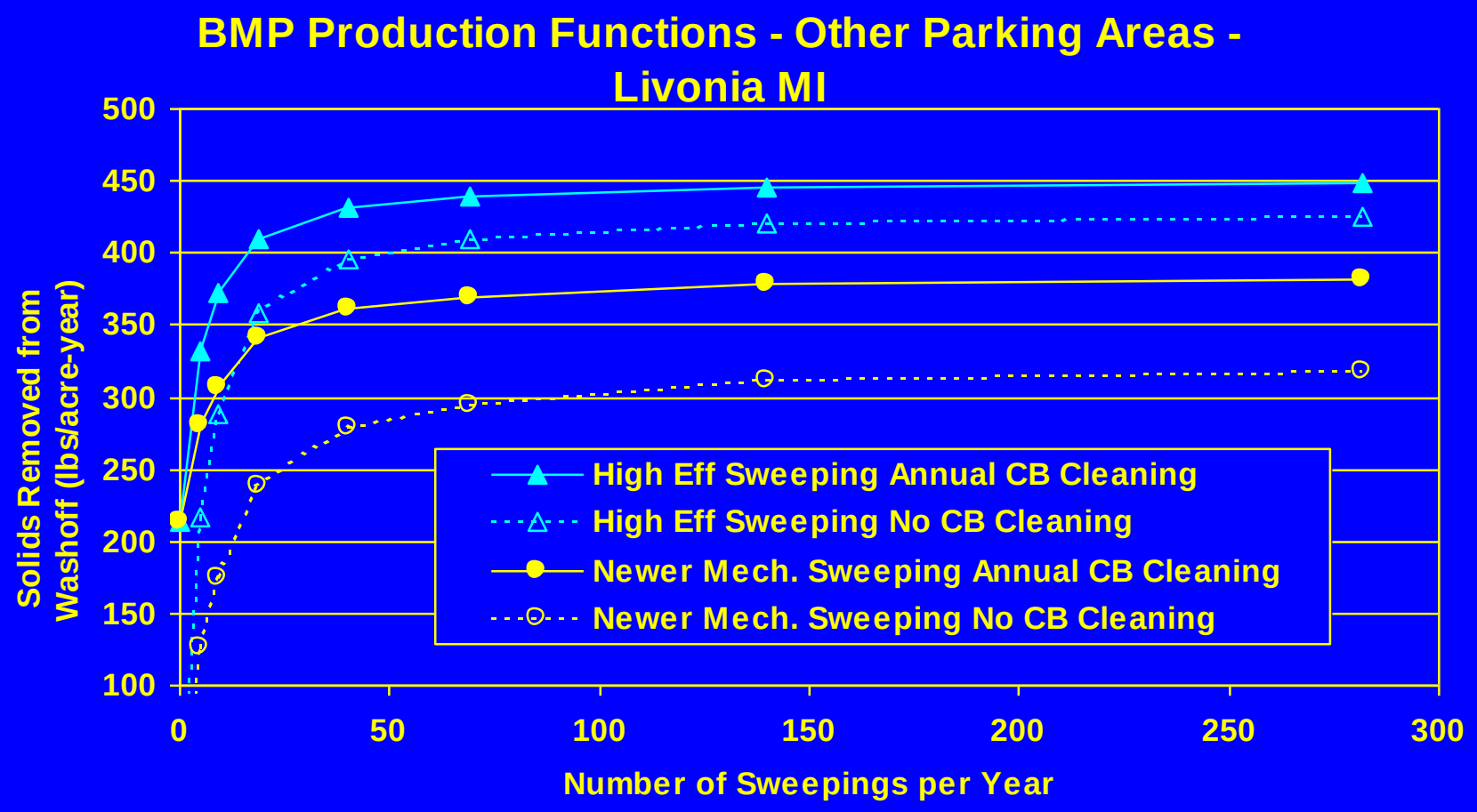
Average Year Rainfall

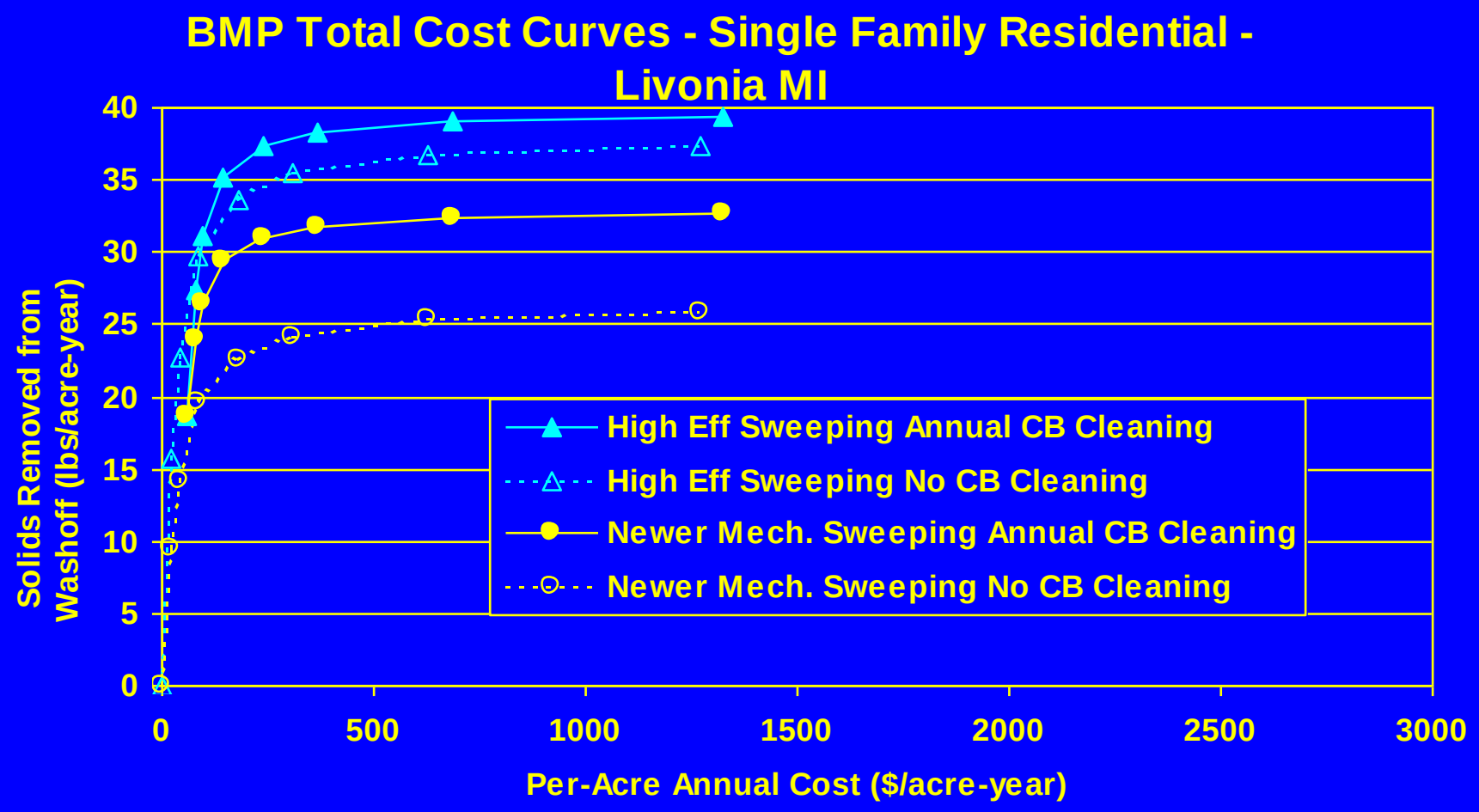
- Used 1961-1998 hourly data from SEMCOG gauge
- Used the RAINEV program to locate and quantify runoff producing rainfall events
- See Volume 6, Chapter 18, p 361 of Toronto Conference Series for description of average year creation
- Excluded average freeze up period of Dec. 21 to Feb. 21
- 61 runoff producing events from Feb. 21 to Dec. 20
- 630 mm total rainfall over 387 hours
- Average event characteristics
  - 10.4 mm depth
  - 6.35 hours duration
  - 1.63 mm/hr intensity

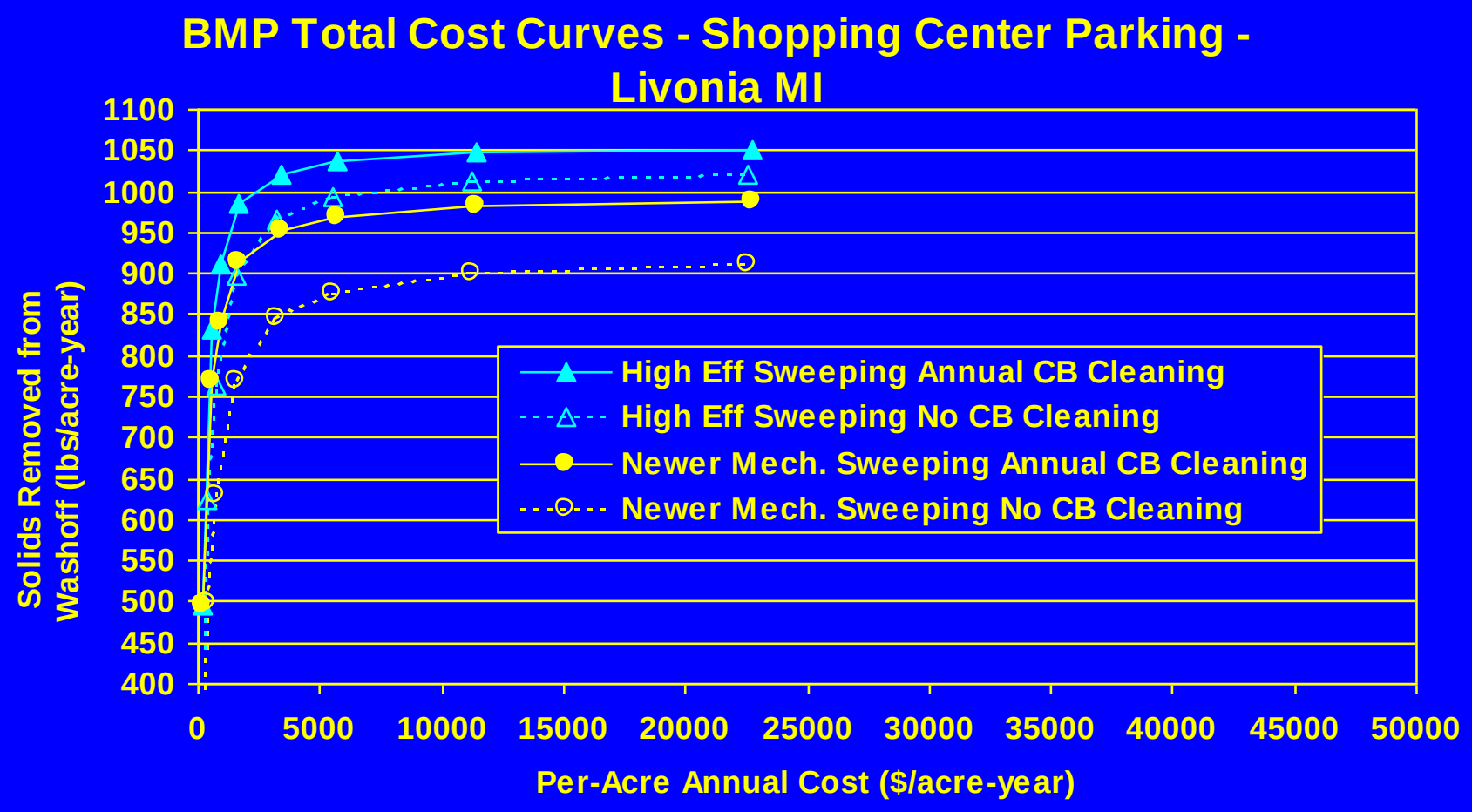

Monthy Rainfall Depths
Livonia, Michigan (1961-1998 Data)
Monthly Total Depth (inches)
0
1
2
3
4
5
6
Rep. Year Depth
Average Depth
1962 Recorded
Jan (87)
Feb (68)
Mar (63)
Apr (86)
May (85)
Jun (78)
Jul (88)
Aug (81)
Sep (76)
Oct (97)
Nov (75)
Dec (64)
Month


BMP Production Functions - Single Family Residential - Livonia MI
Solids Removed from Washoff (lbs/acre-year)
40
35
30
25
20
15
10
5
0
0
50
100
150
200
250
300
Number of Sweepings per Year
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
Newer Mech. Sweeping No CB Cleaning


BMP Production Functions - Shopping Center Parking - Livonia MI
Solids Removed from Washoff (lbs/acre-year)
1100
1050
1000
950
900
850
800
750
700
650
600
550
500
450
400
0
50
100
150
200
250
300
Number of Sweepings per Year
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Tandem Sweeping Annual CB Cleaning
Tandem Sweeping No CB Cleaning


BMP Production Functions - Other Parking Areas - Livonia MI
Solids Removed from Washoff (lbs/acre-year)
500
450
400
350
300
250
200
150
100
0
50
100
150
200
250
300
Number of Sweepings per Year
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
Newer Mech. Sweeping No CB Cleaning


BMP Total Cost Curves - Single Family Residential - Livonia MI
Solids Removed from Washoff (lbs/acre-year)
40
35
30
25
20
15
10
5
0
0
500
1000
1500
2000
2500
3000
Per-Acre Annual Cost ($/acre-year)
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
Newer Mech. Sweeping No CB Cleaning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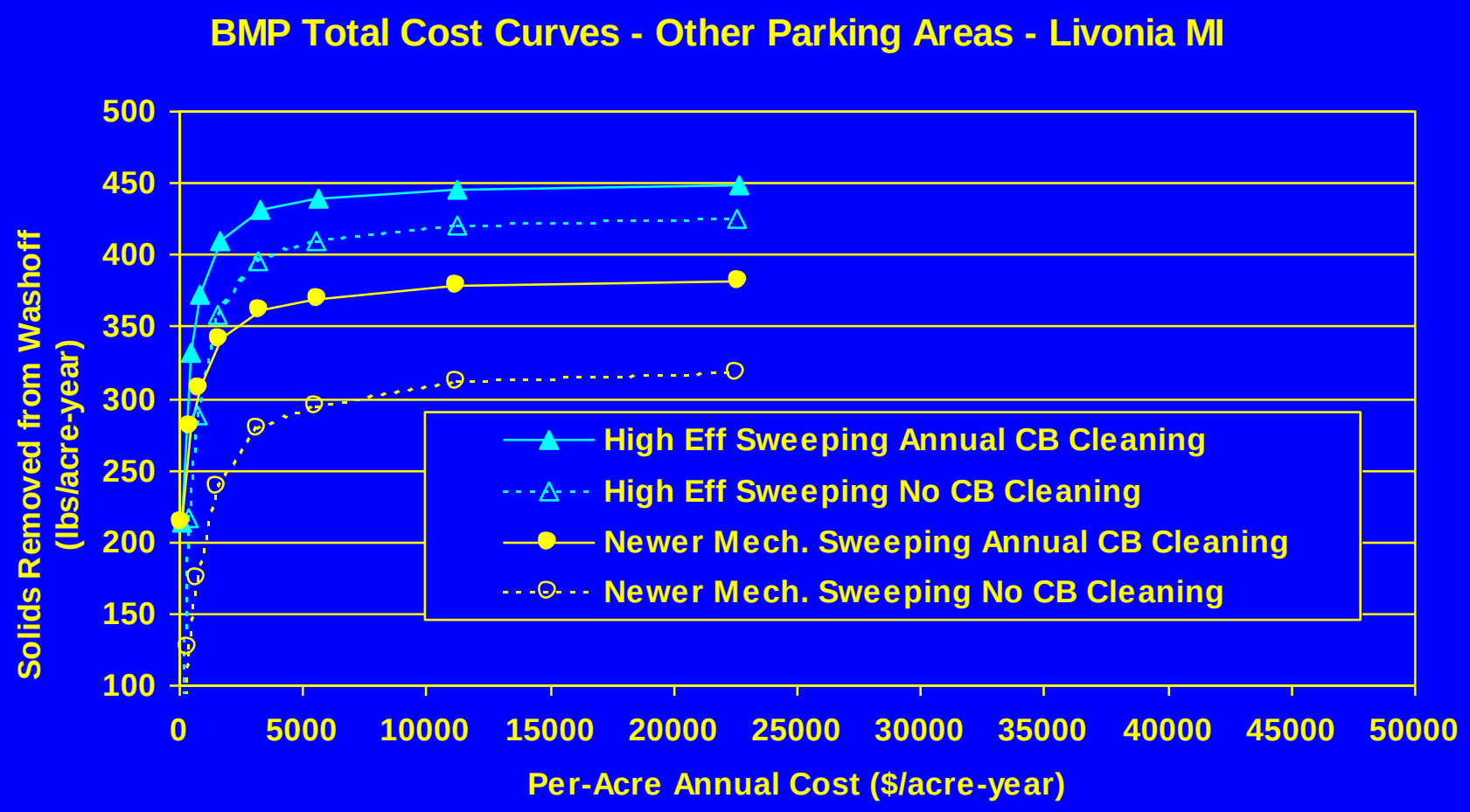
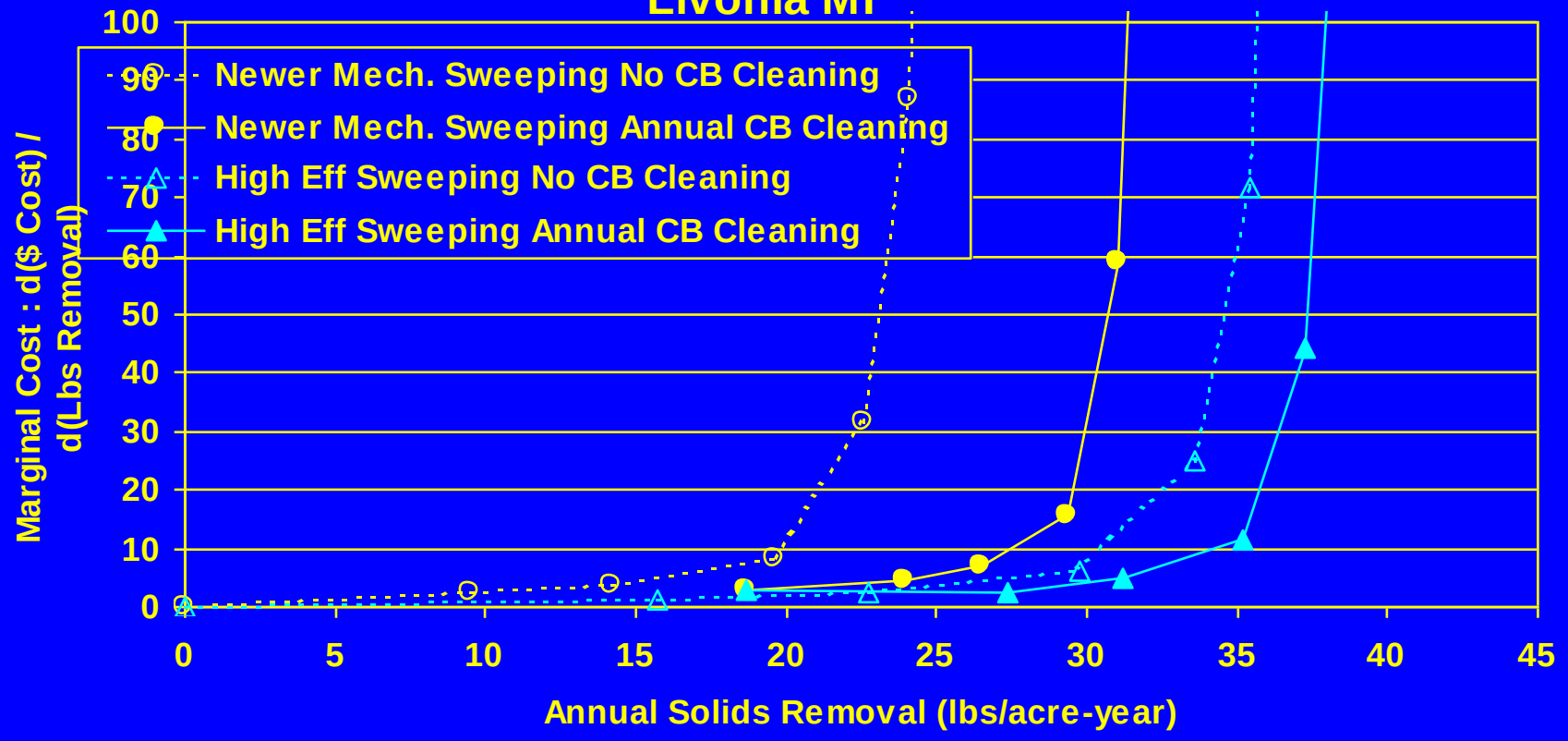
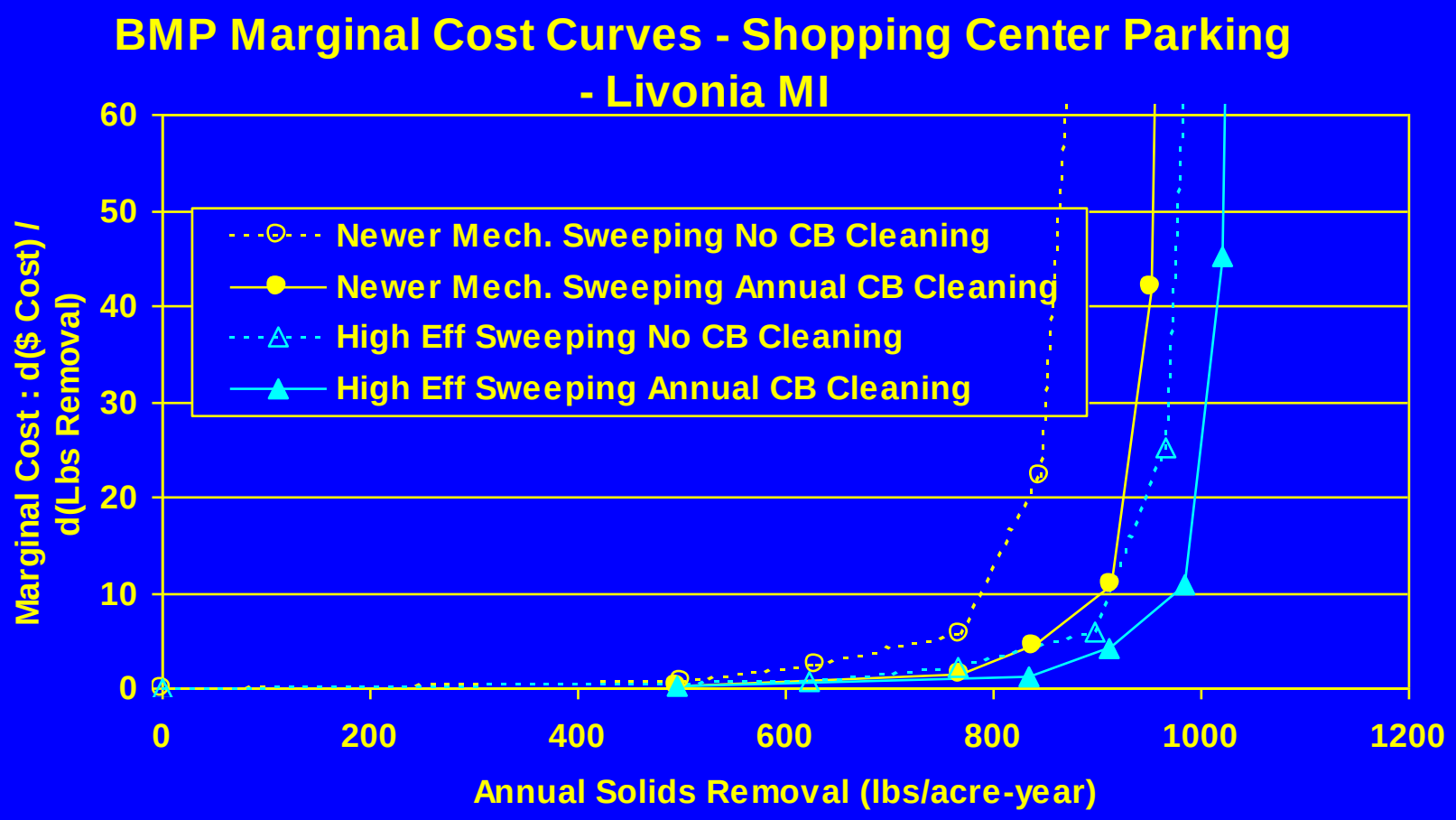
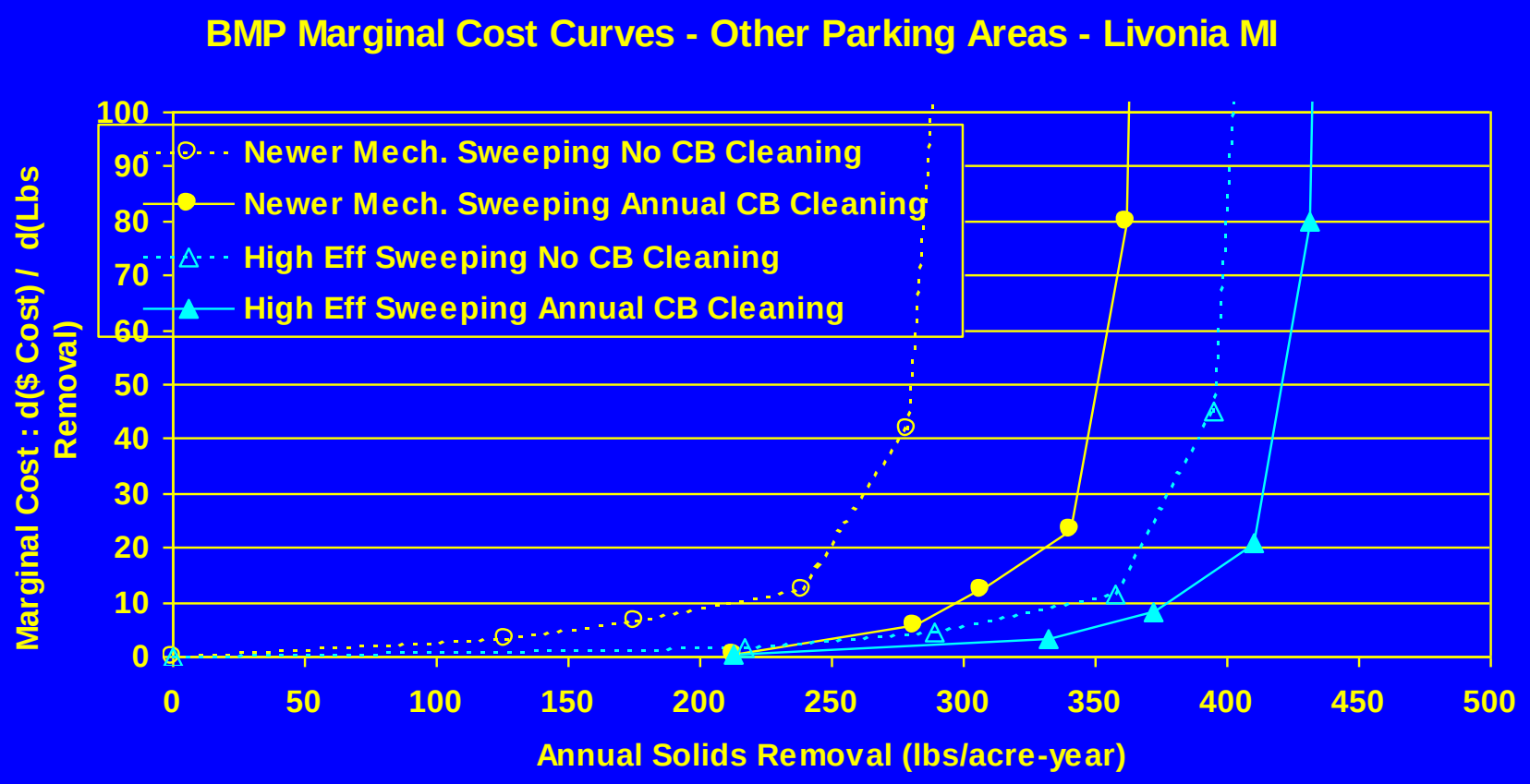
Solids Removed from Washoff (lbs/acre-year)
1100
1050
1000
950
900
850
800
750
700
650
600
550
500
450
400
0
5000
10000
15000
20000
25000
30000
35000
40000
45000
50000
Per-Acre Annual Cost ($/acre-year)
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
Newer Mech. Sweeping No CB Cleaning


BMP Total Cost Curves - Other Parking Areas - Livonia MI
Solids Removed from Washoff (lbs/acre-year)
500
450
400
350
300
250
200
150
100
0
5000
10000
15000
20000
25000
30000
35000
40000
45000
50000
Per-Acre Annual Cost ($/acre-year)
High Eff Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
Newer Mech. Sweeping No CB Cleaning

BMP Marginal Cost Curves - Single Family Residential - Livonia MI




BMP Marginal Cost Curves - Shopping Center Parking - Livonia MI
Newer Mech. Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
High Eff Sweeping Annual CB Cleaning
Marginal Cost : d($ Cost) / d(Lbs Removal)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
Annual Solids Removal (lbs/acre-year)


BMP Marginal Cost Curves - Other Parking Areas - Livonia MI
Newer Mech. Sweeping No CB Cleaning
Newer Mech. Sweeping Annual CB Cleaning
High Eff Sweeping No CB Cleaning
High Eff Sweeping Annual CB Cleaning
Marginal Cost : d($ Cost) / d(Lbs Removal)
Annual Solids Removal (lbs/acre-year)
0
10
20
30
40
50
60
70
80
90
100
0
50
100
150
200
250
300
350
400
450
500

Optimal Effort Levels of Various BMPS on Single-Family Residential Areas

BMP Description	Optimal Level of Effort	Sweepings Per Year	Solids Removed From Washoff Annually (kg/ha)	% Reduction
High-efficiency sweeping & annual CBC	every 15 days	19	39.9	84
Regenerative air sweeping & annual CBC	every 15 days	19	38.7	81
High-efficiency sweeping & no CBC	every 15 days	19	33.8	71
Tandem sweeping & annual CBC	every 30 days	9	34.1	72
Regenerative air sweeping & no CBC	every 15 days	19	31.6	66
Mechanical sweeping & annual CBC	every 30 days	9	30.0	63
Tandem sweeping & no CBC	every 30 days	9	23.4	49
Mechanical sweeping & no CBC	every 15 days	19	22.2	47

CBC – catchbasin cleaning

Annual Pollutant Washoffs with Existing Maintenance Practices

Land Use and BMP Description		Runoff (m/yr)	TSS (kg/ha/yr)	COD (kg/ha/yr)	TP (kg/ha/yr)	Cu (kg/ha/yr)	Zn (kg/ha/yr)
C – Commercial Parking Areas							
Existing Practice	No sweeping or inlet cleaning	0.50	1236	8.01	3.49E-02	3.81E-02	1.24E-01
P – Recreational Parking Areas							
Existing Practice	No sweeping or inlet cleaning	0.50	541	3.37	1.49E-02	1.60E-02	5.00E-02
S – Residential Areas							
Worst Case	No sweeping or inlet cleaning	0.17	47.6	0.31	1.34E-03	1.58E-03	5.21E-03
Existing Practice	Mechanical sweeping (4 times)	0.17	37.9	0.25	1.06E-03	1.27E-03	4.22E-03
Alternative Existing	Regenerative air sweeping (4 times)	0.17	32.1	0.20	8.98E-04	1.09E-03	3.66E-03

Conclusions

- SIMPTM provided reasonably accurate estimates
- Existing levels of residential sweeping are reducing pollutant washoffs by 19 to 33% depending on the type of equipment used
- Pollutant washoff reductions associated with existing levels of catchbasin cleaning were estimated at ~5%
- Street sweeping was more cost effective than catchbasin cleaning
- Mass per unit area annual pollutant washoffs were 3.4 to 7.8 times higher for parking lots compared to residential streets

Recommendations

- City should double the sweeping effort for all residential streets to 8 times from April to November using regenerative air equipment
- City should investigate an ordinance to require regular street sweeping and catchbasin cleaning on privately-owned streets and parking lots
- Conduct sediment accumulation monitoring on additional sites throughout the City including industrial and major arterial roads
- Conduct an in-depth study of long-term equipment capital, life cycle and maintenance costs
- Confirm optimal frequencies before full implementation